AI-Enhanced Analysis of Zurich-Canton Rental Listings

Which factors — size, room count, location, and amenity features — significantly influence monthly rent in Zurich canton?

Group 6

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Scientific Programming — FS 2026

Background, problem, objective

Why we built a Swiss-rental analytics pipeline

- Background — Zurich canton has one of the most expensive rental markets in Europe; rent is the largest line in most household budgets.
- Problem — no clean, structured, canton-wide dataset exists; listing portals expose only human-readable HTML and inconsistent prices.
- Objective — build a reproducible end-to-end pipeline that scrapes, cleans, validates, enriches (LLM) and statistically tests Zurich rentals.
- Research question — which factors (size, rooms, location, features) significantly influence monthly rent in Zurich canton?
- Approach — hypothesis testing rather than black-box prediction; all inputs are auditable through a Streamlit dashboard.

What we used

Public Flatfox API + Python data stack

- Data source — flatfox.ch public JSON API; only Swiss portal not behind Cloudflare. 200 ZH apartments retained after strict filters.
- Filters — canton = ZH, full apartments only, monthly rent unit, 20 ≤ m^2 and CHF 500–15 000; excludes garages, shared flats, commercial.
- Python stack — requests + BeautifulSoup (scrape), pandas + scipy (analysis), OpenAI gpt-4o-mini (feature extraction), SQLite (persistence).
- Visualisation — matplotlib for the printed report figures, Plotly inside Streamlit for the interactive multi-tab dashboard.
- Reproducibility — pinned requirements.txt, .env-based config, deterministic content-keyed LLM cache (no duplicate API calls).

End-to-end pipeline

Eight stages, all idempotent, full chain in ~2 min on warm cache

collect_zurich	→	Flatfox JSON API → raw ZH listings + provenance fields
fetch_details	→	Threaded scrape of each detail page (BeautifulSoup)
cleaning	→	Regex normalisation, typed fields, balcony / parking flags
llm_helper	→	OpenAI feature extraction (gpt-4o-mini, SQLite-cached)
data_quality	→	Validate, dedupe, price QC report, outlier handling
database	→	SQLite persistence + canned SQL queries (NTILE, RANK, AVG)
statistics	→	7 tests — Pearson, Spearman, OLS, Welch, MWU, Shapiro
insights → UI	→	Auto business findings → Streamlit dashboard + PDF figures

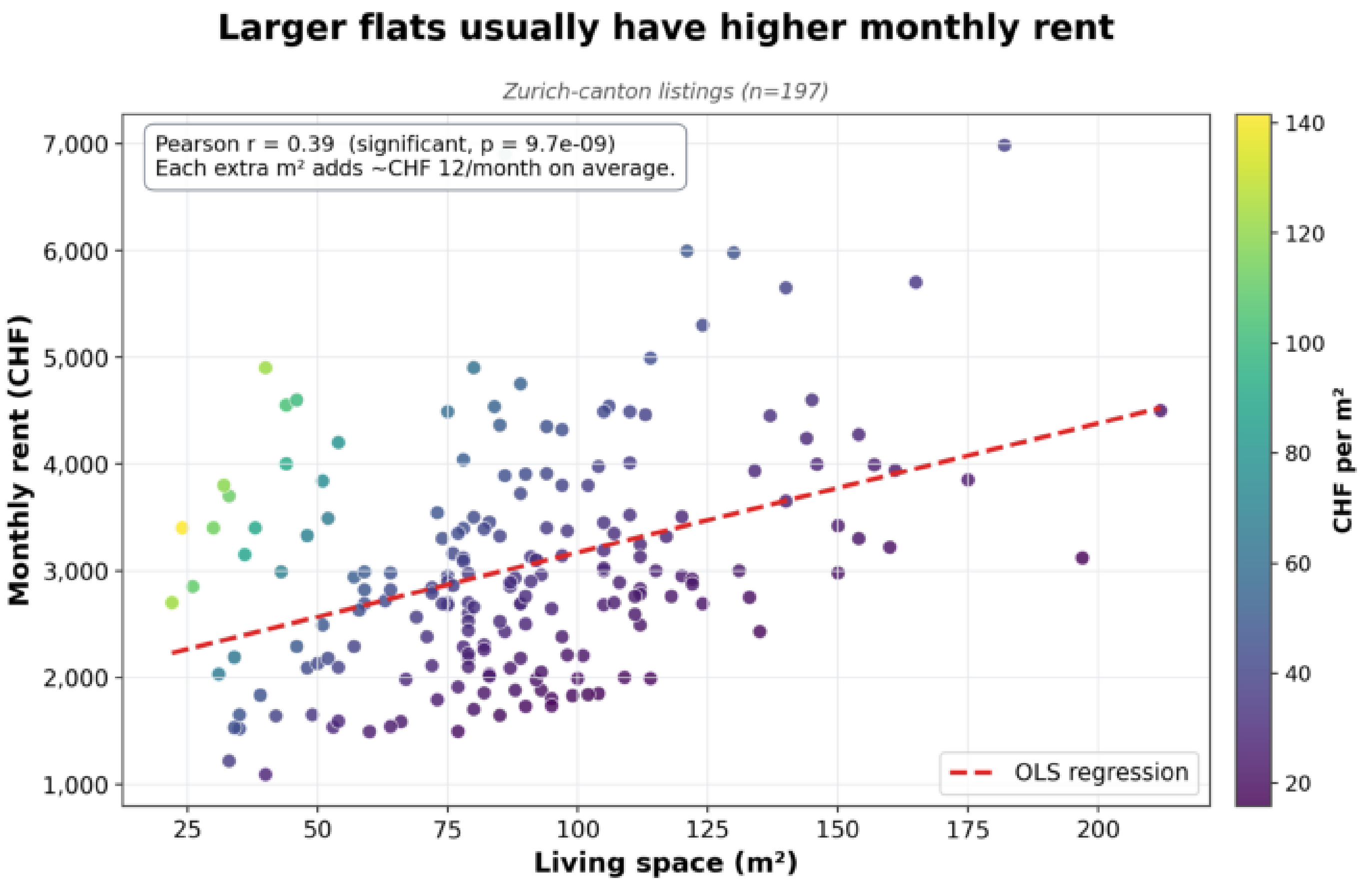
Defensible numbers, not silent failures

Validation, suspicious-price audit, console QC report

- Plausibility bounds — CHF 500–15 000 / month, 15–400 m^2, 1–8 rooms, CHF/m^2 $\in [10, 200]$; non-numeric prices rejected.
- Dedup — natural key (title, price, size, rooms); also filters Tukey $k = 3$ statistical outliers on price.
- Suspicious-price audit — collector keeps raw title text and structured rent_gross; validator flags listings whose gap $\geq$ CHF 100.
- Why it matters — Flatfox occasionally serves stale public_title; without this audit, charts silently use the wrong number.
- QC console report — scraped 200 → valid 197 → suspicious 0 → missing 0 → removed 3 (statistical outliers).

Larger flats usually have higher monthly rent

Pearson r = 0.39 · slope ≈ CHF 12 / m² / month · n = 197



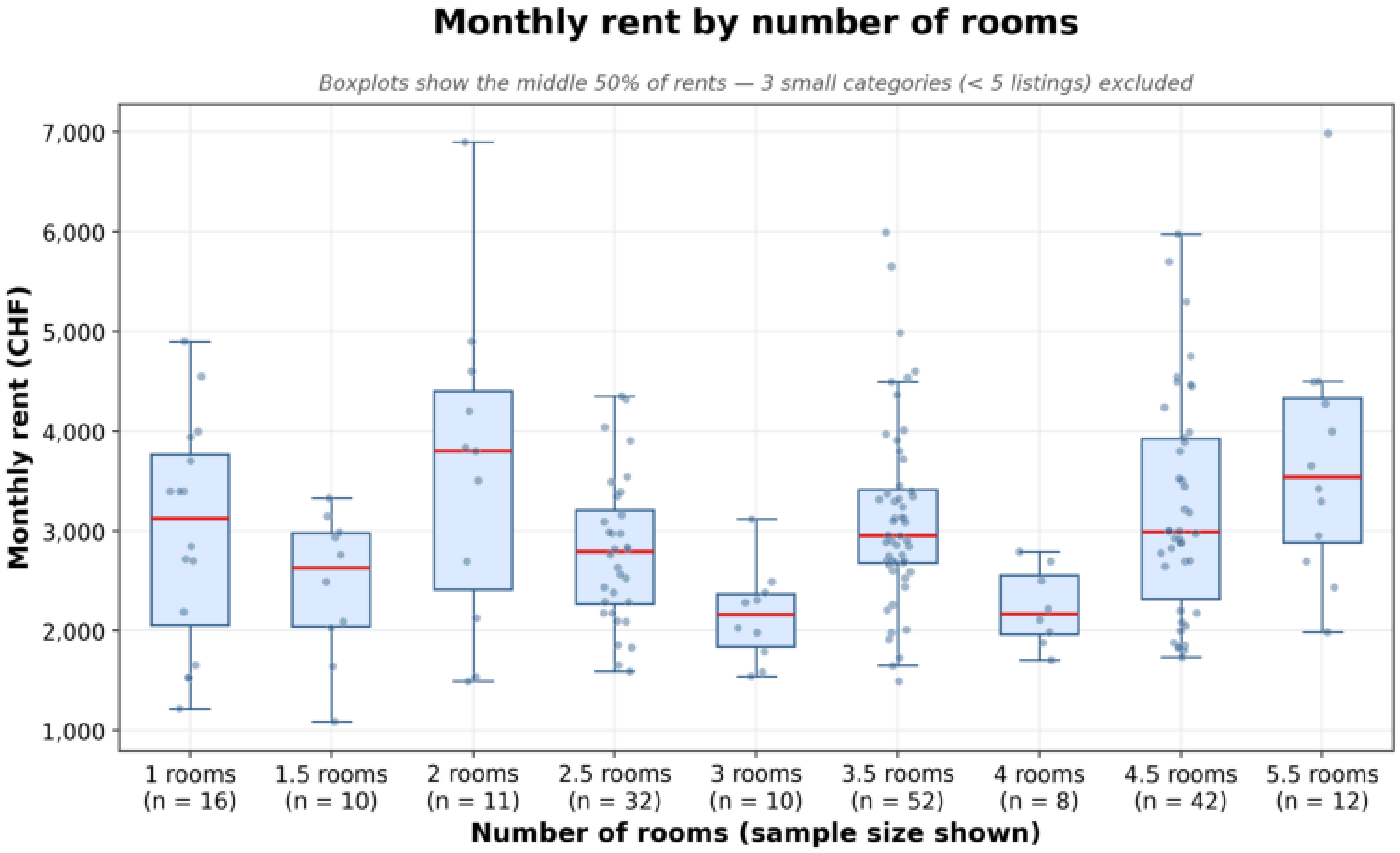
Take-away

Size is the single strongest linear predictor of monthly rent.

But spread is large: same-size flats can differ by CHF 2 000+ depending on location and amenities.

Monthly rent by number of rooms

Boxplot of price per room category (≥ 5 listings each)



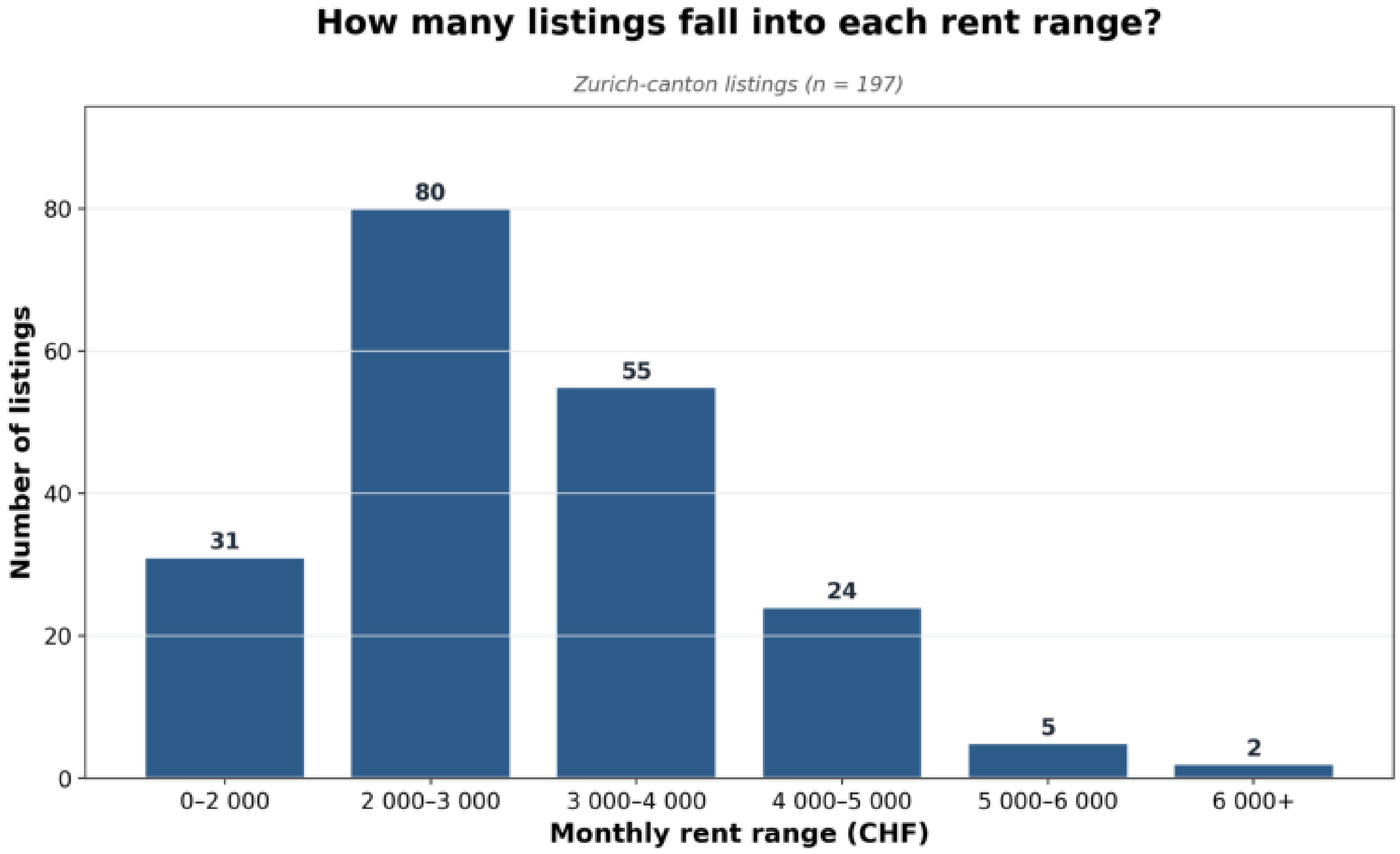
Take-away

Room count is a much weaker price signal than m² (Spearman ρ ≈ 0.15).

Same-room-count flats vary widely in m², which dilutes the signal.

How many listings fall into each rent range?

Fixed bins: 0–2 000, 2 000–3 000, … 6 000+



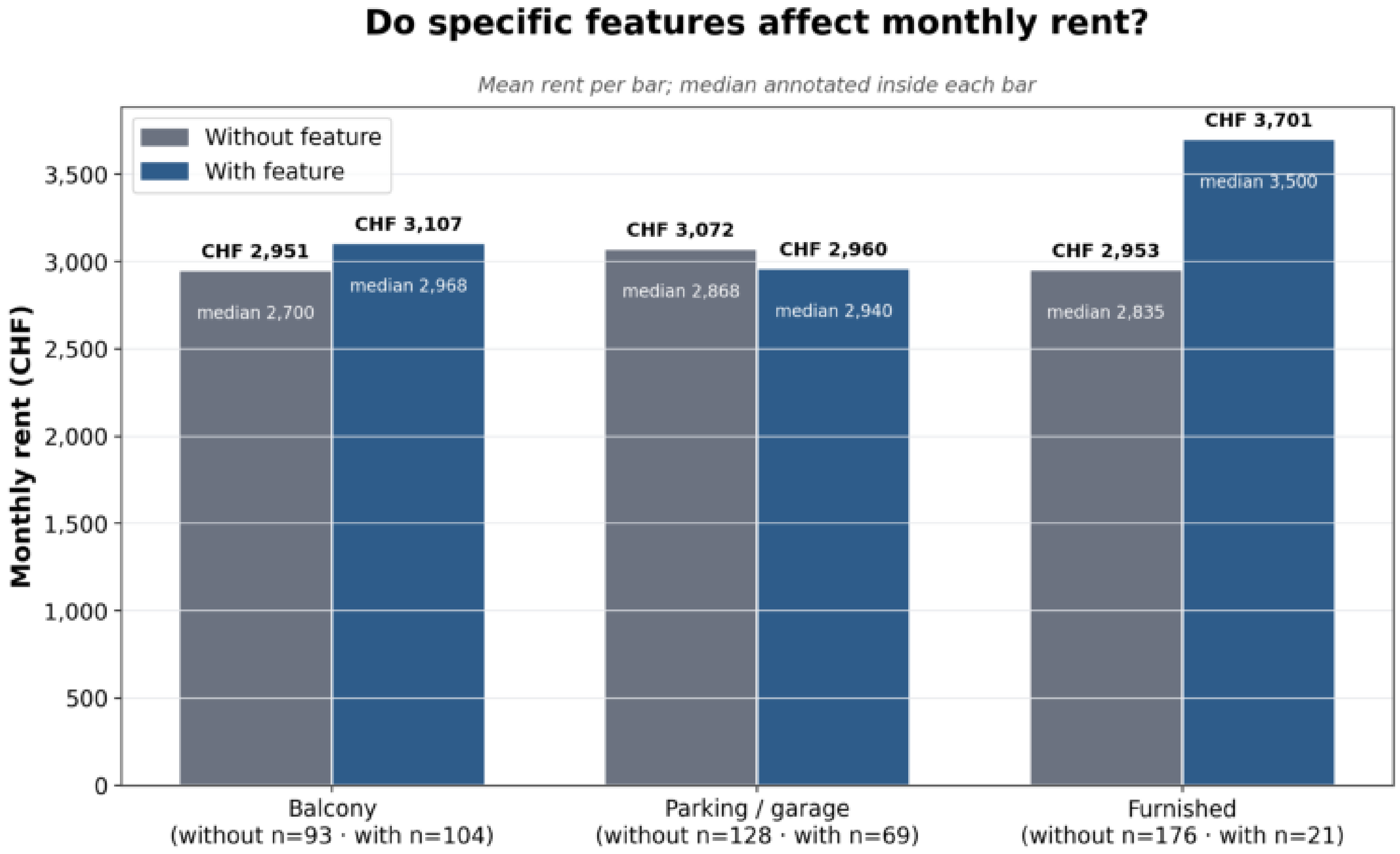
Take-away

Modal bucket is CHF 2 000–3 000 (80 listings).

~70% of Zurich-canton listings rent between CHF 2 000 and 4 000 / month. Only 7 cross the CHF 6 000 line.

Do specific features affect monthly rent?

Mean + median rent, with sample sizes



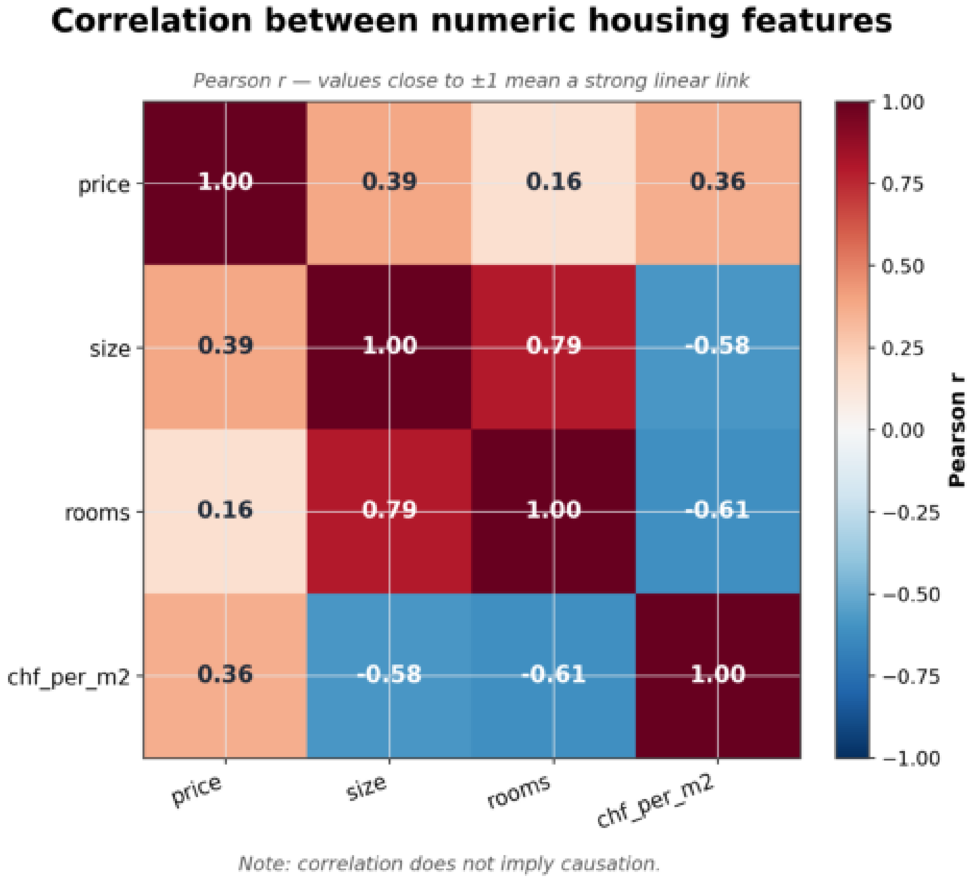
Take-away

Furnished is the only feature with a large effect (+CHF 750, ~25%).

Balcony +CHF 150 not significant. Parking shows a small negative gap, driven by location confounding.

Correlation between numeric housing features

Pearson r — values close to ±1 mean a strong linear link



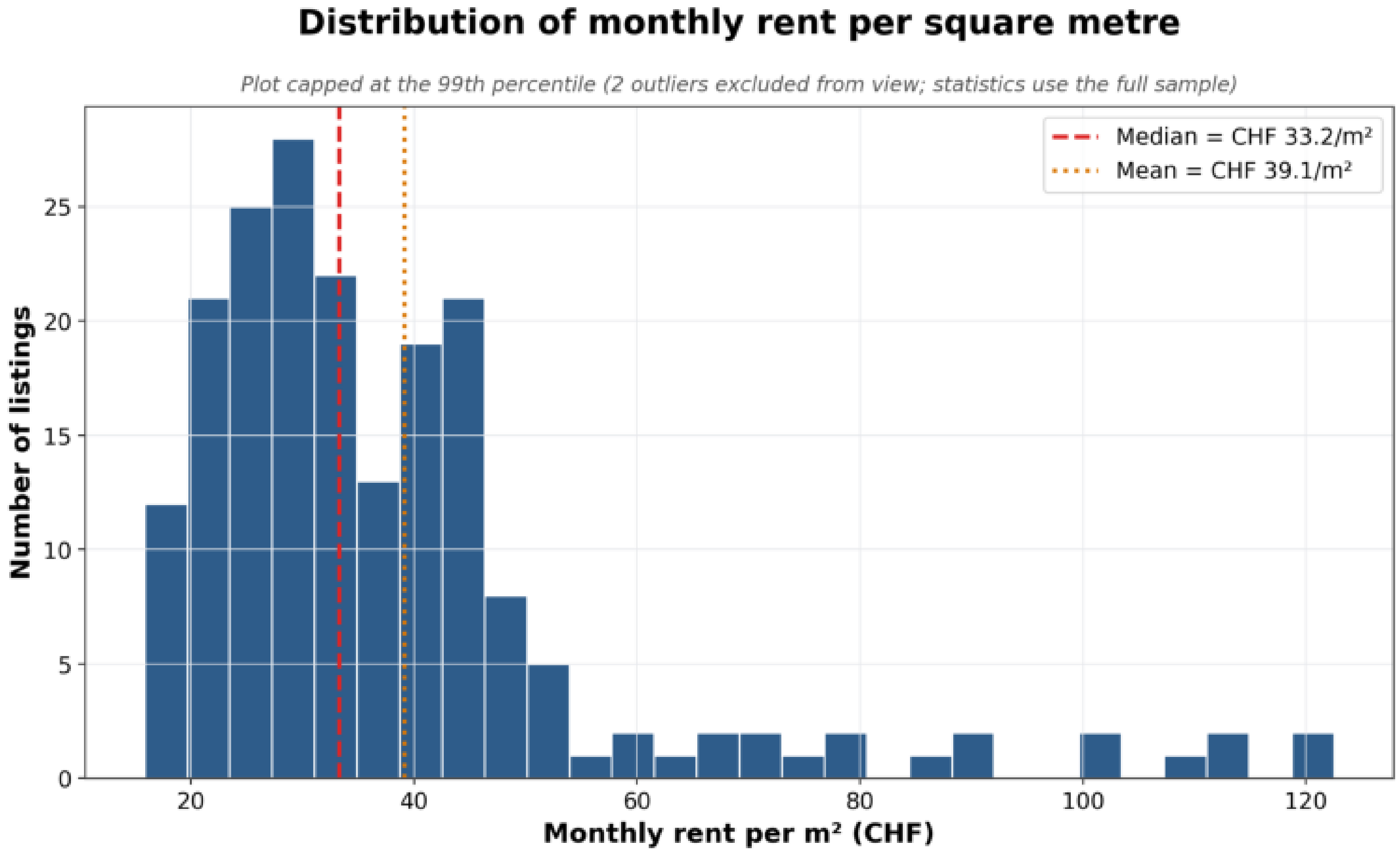
Take-away

Strongest correlations:
- size ↔ price: +0.39
- size ↔ rooms: +0.79
- size ↔ CHF/m²: −0.58

Small flats charge a per-m² premium.

Distribution of monthly rent per square metre

Median = CHF 33.2/m² · Mean = CHF 39.1/m² · right-skewed



Take-away

Median (33.2) < Mean (39.1) → right-skewed distribution.

A small tail of luxury listings pulls the mean upward; the typical listing sits around CHF 33/m².

What the data tells us — and what it doesn't

Interpretation + honest limitations

- Findings — living space is the single strongest linear predictor of rent; ~CHF 12/m² per month. Rooms is a noisy proxy by comparison.
- Amenities — furnished is the only feature with a clear signal (+CHF 750 mean, ~25%); balcony and parking effects are within noise.
- Small-flat premium — size ↔ CHF/m² correlation is −0.58; micro-units consistently extract a price-per-area markup.
- Limitation — Flatfox-only sample of 200 listings; Homegate, ImmoScout and newhome sit behind Cloudflare.
- Limitation — snapshot in time; rents move with season and mortgage rates. A longitudinal study would be a natural follow-up.
- Limitation — LLM feature extraction is fast/cheap but imperfect; validated against a regex fallback on unambiguous cases.

What we delivered

Reproducible pipeline + defensible answers

- Reproducible end-to-end pipeline turns Flatfox listings into a validated, statistically-tested dataset with full audit trail.
- Research question answered — living space dominates rent; room count is a noisy proxy; only furnishing carries a clear premium.
- Engineering contribution — price-QC report cross-checks scraped prices against the listing's own title text and flags mismatches.
- Outputs — Streamlit dashboard (interactive), six light-theme PDF figures (printed report), SQLite DB (downstream queries).
- Future work — multi-canton expansion, time-series snapshots, LLM short-circuit when regex is already confident, external centroid CSV.

Thank you

Questions?

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Source code & figures: GitHub branch feat/price-validation-report-figures

Where everything lives

For graders who want to inspect the code

- GitHub branch — feat/price-validation-report-figures
- Pipeline entry-point — `python -m src.pipeline` (prints QC report)
- Dashboard — `streamlit run app/streamlit_app.py`
- Report figures — `python -m src.report_figures` → reports/figures/
- Validation rules — src/data_quality.py (suspicious-price audit)
- Project context — CLAUDE.md, HOW_TO_RUN.md, README.md